LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
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Objective	Dataset	Model
Predict Outcome: 0 = No diabetes, 1 = Diabetes.	Pima Indians Diabetes Database from Kaggle; file: diabetes.csv (768 rows, 8 inputs).	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV → inspect data → replace invalid zeros → train-test split (80/20, stratified, random_state = 42) → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.813	Overall class-separation ability

Confusion Matrix (test set = 154 patients)

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

Key Findings

The model obtains a ROC-AUC of about 0.813 and detects approximately 70.4% of the positive cases in the fixed test split (38 of 54). Glucose (coefficient 1.18) and BMI (0.71) have the strongest positive model coefficients, followed by Pregnancies, DiabetesPedigreeFunction and Age. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases.

Limitation: This small historical dataset represents a specific population. The model is suitable for learning classification concepts, not for medical diagnosis.

Final Conclusion

In this project I trained a balanced Logistic Regression model to predict diabetes. Invalid zeros in five medical columns were treated as missing values and filled with the training-set median, then the features were scaled. The model separates the two classes reasonably well (ROC-AUC 0.813) and catches about 70% of real diabetes cases. The trade-off is a modest precision of 0.603, meaning some healthy people are flagged by mistake, which is acceptable for screening where missing a patient is worse than a false alarm. Glucose and BMI turned out to be the most influential factors. Through this work I learned how data cleaning, scaling and class balancing affect a classifier, and why accuracy alone is not enough to judge a model.

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